Deep-Learning-Enabled Automatic Optical Inspection for Module-Level Defects in LCD

Haidi Zhu, Jingchang Huang, Huawei Liu, Qianwei Zhou, Jianqing Zhu, *Member, IEEE*, and Baoqing Li

***Abstract*—Liquid crystal display (LCD) defects detection on module level is increasingly important for flat-panel displays (FPD) industry to increase the production capacity via machine vision technology. However, it is an overwhelmingly challenging issue due to various difficulties. This article discloses a practical automatic optical inspection (AOI) system consisting of hardware structure and software algorithm to detect module-level defects. The AOI system is the core component to build a distributed integrated inspection system with the help of the Internet of Things (IoT). Starting from the analysis of the challenges encountered in module-level defects inspection, a delicate photograph scheme is proposed to reveal different kinds of defects. In order to robustly work on the module-level defects detection with complex situations, a novel framework based on YOLOV3 detection unit is proposed in this article, including the preprocessing module, detection module, defects definition module, and interferences elimination module. To the best of our knowledge, this is the first work that designs a practical AOI system for module-level defects detection. In order to demonstrate the effectiveness of the proposed method, extensive experiments have been conducted on the manufacturing lines. The evaluation of the detection performance of the AOI system in comparison with a manual scheme indicates that the proposed system is practical for module-level defects detection. Currently, the proposed system has been deployed in a real-world LCD manufacturing line from a major player in the world.**

***Index Terms*—Automatic optical inspection (AOI), deep learning, liquid crystal display (LCD), smart manufacturing.**

I. Introduction

IN THE past few years, the flat-panel displays (FPD) industry is developing quickly since the enormous demands for electronics devices, such as smartphones, laptops, monitors, and other portable electronic products, in which FPDs are indispensable components. In general, the FPD can be categorized into four main types, including liquid crystal display (LCD), plasma display panel (PDP), organic light-emitting diode (OLED) display, and polymer light-emitting diode (PLED) display. Currently, LCD technology dominates the FPD market while thin film transistor liquid crystal display (TFT-LCD) is the most popular type of FPD due to its light weight, low-power consumption, and full-color display capabilities.

As shown in Fig. 1, the TFT-LCD manufacturing process can be divided into three stages, including TFT array process, cell process, and module process. Each stage involves several subprocesses [1]–[5]. In fact, the manufacturing pipeline is exceedingly cumbersome and complex in the sense that every subprocess inevitably brings about various defects that finally hurt the yield. In order to pick out defective panels from the production lines as early as possible to avoid the waste of materials in later processes, experienced operators are employed to select products in question through visual inspection methods at each subprocess station. However, the manual solution of defects detection is primarily dependent on personal experience and subjective judgments. Accordingly, the consistency of inspection standards is an issue that poses a diverse imapct on products' quality and reputation. In order to avoid misjudgments caused by human eyes and decrease labor force consumption, automatic optical inspection (AOI) technique is strongly demanded in manufacturing industry. It has become one of the most active research areas. Although AOI equipment has been widely deployed in TFT-LCD array process and cell process [11], [30], [31] in practice, AOI solutions for the module-level detection are blank since an army of challenges are encountered in this stage, specifically as follows.

1) *Diverse Defect Types:* The module process is the last phase of the TFT-LCD manufacturing. The defects missed by the previous inspection processes are accumulated in this stage. In theory, all kinds of defects appeared in previous stages can be encountered in the module-level process station. From the first row of Fig. 2, some of the complexities of defects can be glimpsed.
2) *Super Large Dynamic Range of the Defect Size:* Defect types with smallest size basically include "bright spot" and "dark spot." Both are dot level, which are defined as defects once the abnormal region exceeds the (1/2) dot area (the size of a dot is 20×60 um in our project).

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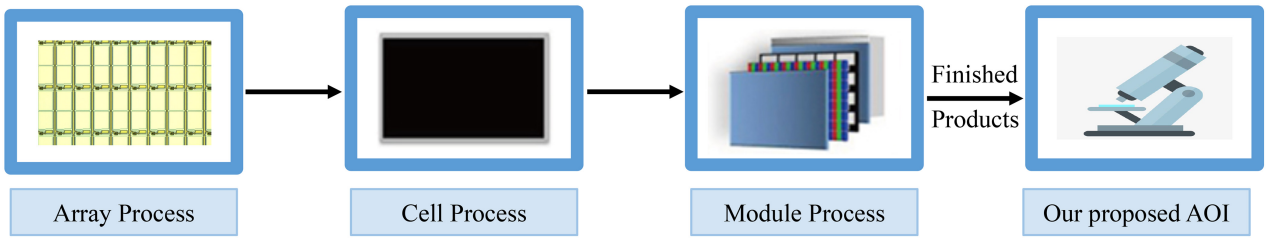


Fig. 1. Manufacturing process.

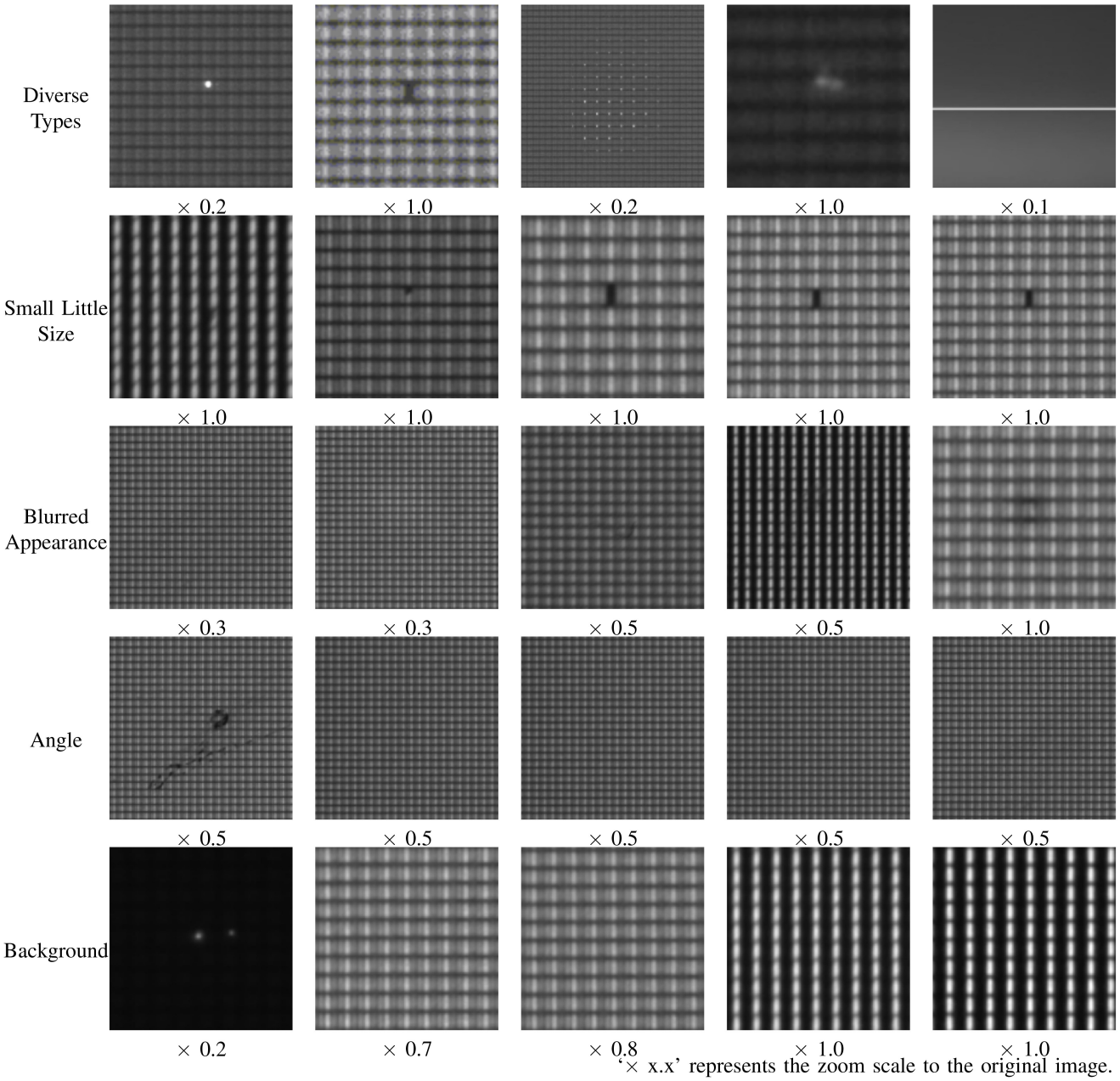


Fig. 2. Overview of defect characteristics.

In contrast, defects with large size are likely cover the whole panel. Several defects with small size are shown in the second row of Fig. 2.

3) *Blurred Appearance:* As shown in the third row of Fig. 2, some defect types in module level, which are missed in the previous inspection stages and covered by materials conducted in later processes, are hard to be clearly observed. Hence, it is hard to detect these blurred defects, which locate in the bottom layer of LCD.

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4) *Varied Visible Observation Angles:* It is known to all that different kinds of defects are caused by different physical reasons and occur in different layers of the LCD panel with dozens of geometric shapes (polyhedron, sphere, or strip). Different geometries have different capability in reflecting light at diverse visible angles and directions. For example, some defects are visible in the main view while the others are only observed in slide views. Therefore, images inputted to the module-level AOI system should be captured at different angles. As shown in the fourth row of Fig. 2, defects can be noticed from a special observation angle while cannot from other four angles.
5) *Diverse Visible Backgrounds:* In practice, it is found that a given defect is only visible with some certain kinds of background images for the various contrast of a defect with different background pictures. As shown in the fifth row of Fig. 2, the defect with bright appearance can only be observed in the dark background screen. In order to reveal different kinds of defect types, it is necessary to display background images as more as possible in the panel. However, the more background images display, the more time takes. Therefore, a comprehensive and compact combination of background images should be elaborately designed to build a visual environment with diverse contrast to reveal various defects.
6) *Multidimensional Defect Definition Criteria:* In the tasks of traditional object detection, objects are defined only according to their appearances. However, for defects detection, some special abnormal regions are judged as defects not only according to their appearances but also depend on the size, amount, and even neighbor distance. For example, if the number of dot-level abnormal whose area ranges from $(1/2)$ dot to 1 dot in one panel is less than 3, it will be up to standard. Otherwise, it is considered as the defective product. Furthermore, if the area of one of the abnormal regions exceeds 1 dot, it will be directly regarded as defect.
7) *Numerous Interferences:* The manual assemble operation is necessary after each subprocess of module-level production. It is inevitable to bring about dust interference. In addition, protective films are usually required to be attached on the final version of module-level products. Slight damages of protective film (film damage is not considered as defect) will result in irregular appearances, which are easy to be mistaken for defects.

An ocean of efforts has been made to study module-level defect inspection. However, they only aimed to solve limited defect types [32], rather than a complete solution qualified for all defect types. The traditional methods [33]–[35] cannot capture the features of different kinds of defects simultaneously.

In recent years, deep learning technology achieved great success in object detection [12]–[16], [18], [19], [21]–[29]. Apart from artificial features obtained by traditional algorithms, deep learning algorithms conduct representational learning on a large amount of data. Thus, they are more flexible in practical applications with stronger generalization ability for their scalability. For example, R-CNN [12] proposed by Girshick *et al.* was a successful case of applying deep learning algorithms to object detection. After that, it was further improved by Fast RCNN [13] and Faster RCNN [14], which employed the region proposal network (RPN) to obtain the region of interest (ROI). Although these algorithms provide guidance for industrial product detection, high computing force is indispensable to obtain a satisfactory accuracy. To overcome this problem, Redmon *et al.* [15] proposed a new network structure [you only look once (YOLO)], which is an end-to-end object detection model without using RPN. It gained higher computational efficiency together with satisfied accuracy. Thus, it is improved and applied [15], [22]–[25] widely for its high efficiency.

At present, deep learning technology is becoming more and more popular in industrial fields [31], [32], such as traffic signs classification [7]–[10], industrial product defect detection [30], [36], and mechanical fault diagnosis [37], [38]. Considering the challenges of LCD defects detection mentioned above and the significant identification capability of deep learning algorithms, a practical AOI system is presented for module-level defects inspection in this article. It can be the core of an Internet of Things (IoT) paradigm for the optical inspection tasks. A distributed intelligent inspection system can facilitate dynamic training and updating models. It provides data support for various factory decisions and strengthen information management, which is the benefit of arranging production plans more reasonably. Hence, the work in this article is the significance of detecting the LCD defects with various challenges and promoting the development of the smart factory and Industry 4.0. Specifically, for the AOI system, the LCD panel is triggered first to display multiple background images. Second, multiple cameras are used to capture the LCD images with multiple backgrounds from different angles. Finally, obtained images are fed into a detection system based on the deep learning algorithm. As shown in Fig. 3, the challenges and solutions are stated one by one. Specifically, with a large amount of data including diverse defect types with large dynamic range of defect size, the AOI system based on deep learning methods is naturally more suitable to solve these difficulties. Capturing images with different background from multiple angles can alleviate the problems described in challenges 4) and 5). With a criteria parameter filter module shown in Fig. 7 (the software algorithm framework illustrated in following Section III-A), multidimensional defect definition criteria can be applied to solve the challenge 6). Facing the challenge 7), images are captured in a power-off state as well to detect the dust interference. Specially, a YOLOV3 unit shown in Fig. 7 is designed to detect the film damages. Hence, the film damages are removed from the final detection outcomes. Multiple detectors are designed specially to ensure the detection performance and work in parallel to obtain a high computational efficiency as shown in Fig. 7. In general, the number of defects in one panel is small. It is not necessary to generate too much dense proposals. Numerous interferences are eliminated by special design described above. Utilizing two-stage networks as the detector unit will bring about too much unnecessary resource and runtime consumption. Without

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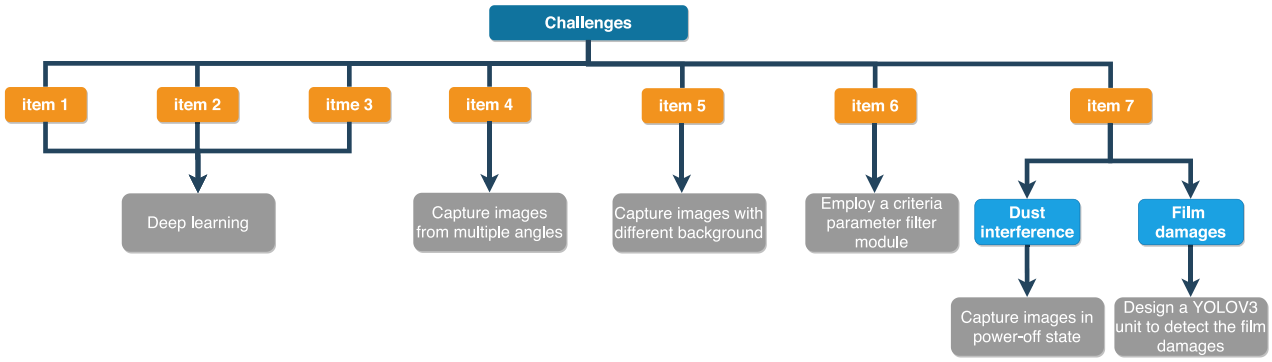


Fig. 3. Each challenge and its corresponding solutions.

a very complex single detector unit, it can also achieve a satisfied detection outcome.

With the application of the artificial intelligence technology, the intelligent manufacturing mode driven by big data has been formed. A new round of industrial revolution (Industry 4.0) [39] characterized by intelligent manufacturing has also been gestated. Smart inspection is an inevitable tendency for defects detection, which is an important component in the smart factory. The AOI system we proposed can be employed for it. Its high accuracy, fast detection speed, advanced intelligent performance, incomparable consistency, and repeatability meet the needs of large-scale online detection. A large amount of physical labor can be saved. Resources and energy can be utilized more effectively, which is significant to realize the green development of the industry.

The main contribution of this article is the design and implementation of an AOI system for detecting the module-level defects in LCD panels. The AOI system is the core of an IoT paradigm for automatic inspection detection. Specifically, the innovation points proposed in this article are summarized in the following.

1) A delicate photograph solution to reveal all kinds of defects is proposed.
2) A preprocessing method is proposed to work well on ultrahigh resolution images for extremely tiny defects detection.
3) A combined detection framework based on YOLOV3 is designed to detect the module-level defects.
4) An interferences elimination method is employed to refine the detection result by suppressing the ill impact of the dust and protective film damages.
5) A distributed intelligent inspection system with the capabilities of automatically collecting defective images is designed and implemented in a well-known manufacturing enterprise.

This article is organized into five sections including the present one. Section II illustrates the principle of defects imaging and overview of the system structure. Section III discusses the proposed detection framework basically, including the preprocessing module, deep convolutional neural network detection unit, interferences removal method, and model composition. In Section IV, the performance of our proposed AOI system is verified by large-scale acceptance test. In Section V, the importance of the AOI system we proposed for intelligent inspection in the IoT paradigm is discussed. Finally, in Section VI, some of the distinctive features of this investigation are highlighted.

II. System Overview

In this section, we elaborate the choice of the background image library. In addition, the hardware architecture of the AOI system is described particularly.

A. Background Screen

It is observed that some defect types are visible with certain background images. For example, "white spot" and "bright dot" are clearly shown on the dark background while "dark spot" and "dark dot" cannot be noticed.

Table I shows that different kinds of defects are observed with certain background images. Although the typical defect types listed in Table I are visible with background images of red, green, and blue, the clarity degree of defects is hardly enough. In order to detect defect types as more as possible, a combination of background screens is considered for visual inspection. In addition, it is necessary to keep the combination as compact as possible to save the time consumption of screen displaying and image capturing. In addition, dust is not easily observed when the panel is triggered. However, it can be captured in the power-off state. The panel with dust is not regarded as defective. It is important to capture a dust image to remove the dust interference. Other interferences, such as film handle and seal, are captured on the gray image. They are eliminated with our framework, which is described in detail in Section III-D. Hence, the background screens employed are dark, white, gray, and power-off.

B. Hardware Architecture

In this article, the proposed AOI system, which is a node of the distributed system in an IoT paradigm, includes the mechanical loading and uploading subsystem, LCD panel trigger subsystem, imaging subsystem, and visual analysis subsystem. The structure of the entire system is shown in Fig. 4. Specifically, the mechanical loading and uploading subsystem is responsible for providing the LCD panel to test and taking it away after finishing the detection

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TABLE I
VISUALIZATION OF DEFECTS ON DIFFERENT BACKGROUND IMAGES

Screen Name	image	Defect types ("√"denotes visible,"×"denotes invisible)							
		White Spot	Dark Spot	Bright Dot	Dark Dot	Cell Anomaly	Surface Anomaly	Subface Anomaly	···
Dark		×	√	×	√	×	√	×	···
Gray		√	√	√	√	√	√	√	···
White		×	√	×	√	×	×	×	···
Red		√	√	√	√	√	√	√	···
Green		√	√	√	√	√	√	√	···
Blue		√	√	√	√	√	√	√	···
Board		√	√	×	×	×	×	×	···
Baby		√	√	×	×	×	×	×	···

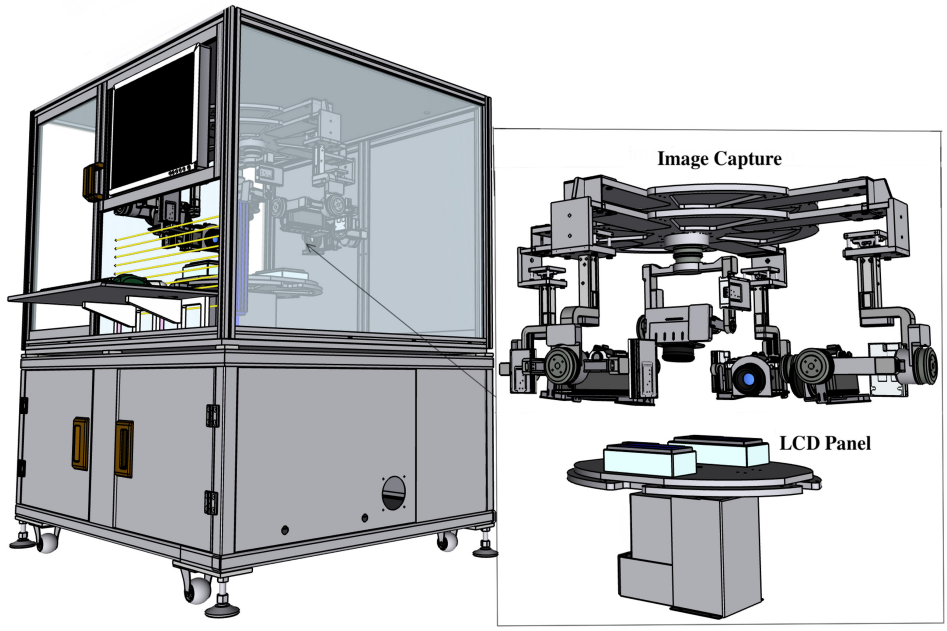


Fig. 4. Hardware architecture of AOI system.

operation. The LCD panel trigger subsystem is in charge of controlling the LCD panel to display different background screens in sequence. Furthermore, an imaging subsystem is required to capture all the LCD screens. Finally, a visual analysis subsystem based on the deep convolutional neural network unit is responsible for detecting the defects.

The LCD panel trigger subsystem and imaging subsystem are further detailed as follows.

1) *LCD Panel Trigger Subsystem:* When an LCD panel is loaded onto the platform, the LCD screen trigger subsystem will turn on the panel and trigger the screen to display the first background image. After that, the imaging subsystem will directly capture the current display. Subsequently, the process will be repeated until all the background images are exhausted.
2) *Imaging Subsystem:* The defects have dozens of geometric shapes (polyhedron, sphere, or strip) locating on

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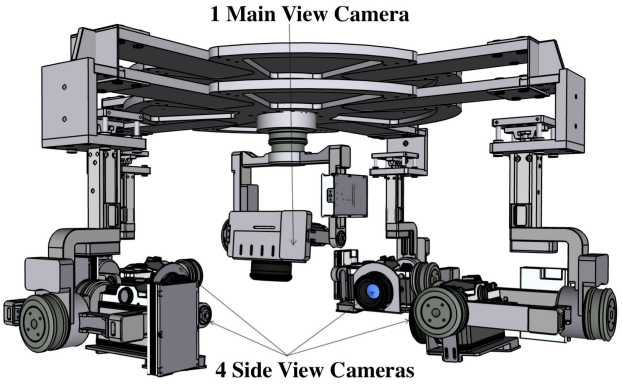


Fig. 5. Imaging subsystem.

different layers of the LCD panel. Therefore, visible angles and directions are diverse. In this article, a total of five cameras are employed to take pictures. One is placed in the vertical direction and the other four are set around the LCD panel in side directions as shown in Fig. 5.

In order to shoot extremely tiny defects, an industrial camera with ultrahigh resolution is necessary. Considering the resolution and the cost, a grayscale camera with 71 million pixels (10000×7096 pixels) is employed. Moreover, we also need to use color cameras since some kinds of defects are colored. Therefore, the main view camera in Fig. 5 is configured as a high-resolution gray camera while the color cameras with 5 million pixels (2560×1920 pixels) are employed on the four side views. In order to trade the detection accuracy off computational efficiency, the main view camera will photograph all the selected background screens while the side view cameras only take pictures for the gray background screen for the best defects visibility in the gray background.

C. Captured Images

The captured images of one panel are shown in Fig. 6, including four images from the main view camera (related white, gray, dark, and power-off) and four images from side view cameras (related four angles). It is observed that the contrast and brightness of each main view image are different and some defects are shown more clearly in color pictures.

III. Proposed Inspection Method

With so many interferences, the real-time detection for diverse defects with high accuracy is an intensely challenging work due to the excessive amount of data. In the following sections, principles and tricks of the detection framework will be detailed.

A. Model Composition

During the process of LCD panel manufacturing, there are many kinds of defects and some are greatly different from others. According to the requirement for defects inspection, more than 95% hitting rate and less than 5% missed rate are necessary. Undoubtedly, a single detector cannot be adequate for such a task with the challenges mentioned above. Therefore, as shown in Fig. 7, the algorithm framework was proposed in this article, in which dozens of defect detectors are needed. Each detector (YOLOV3), which is represented as Detector_XXX in Fig. 7, is responsible for one or several defect types. Specifically, defects with similar appearance share the same detector. Therefore, the number of detectors is up to the similarity between the defects. The submodules in Fig. 7 will be introduced in the following Sections III-B–III-D.

First, images of a panel are preprocessed. It is described in detail in Section III-B. Then, the images are inputted to different detectors according to the capturing angle and background. Specifically, the defect samples are grouped according to their visible backgrounds and inputted into the corresponding detector separately. According to Table I in Section II-A and Fig. 5 in Section II-B, a total of eight detectors are employed in our framework. After training all the detectors, the models are composited to refine a complete detection capacity. One of the detectors is utilized to detect the inference such as film damage. Hence, the inference can be further removed from the detected results with the SideDelete_Inferences and MainDelete_Inferences modules. In addition, a Dust_Binary module can detect the dust with the image of a panel captured under a power-off state according to a threshold. After that, the dust is removed from the results. In addition, CriteriaPara.Filter_XXX modules are used to define the defects. Specially, some filters are embedded in our proposed framework. They are designed to configure different parameters to meet different defect definition criteria. For example, a panel of 7 inches is regarded as a defect when the area of one of its damaged dots is more than (1/2) dot area. It is considered pleasurable until its damage areas surpass (3/4) dot area for an 8-inch panel. Finally, the detected outcomes are achieved by combining the results from the side and main view images. The results of four detectors using four side view cameras are fused according to their IoU [42] with each other. When the IoU is greater than the set threshold, the two results are merged. The same is the fusion of results obtained by the main view camera. Finally, the results of these two parts are fused with the same method.

B. Preprocessing

1) Edge Cutting: The size of the panel varies with product types while the configuration of cameras should not be frequently altered. It is inevitable to capture redundant regions around the panel during the process of taking photographs. In order to be adequate for diverse product types, the edge detection algorithm [41] is employed to cut off the redundant fringe adaptively. Taking the 7-inch panel for example, a real image with size of 9600×5760 pixels is cut from the raw image with 10000×7000 pixels.

2) Sliding Window: After cut as described in Section III-B, the image is still too large (9600×5760). It is hard to train and infer on the image directly using common GPUs. Hence, the image is divided into subimages as shown in Fig. 8, which is inspired by the process of sliding window [12] in the field of signal processing. Specifically, the raw image is separated into 80 subimages and each of which contains 1024×768 pixels.

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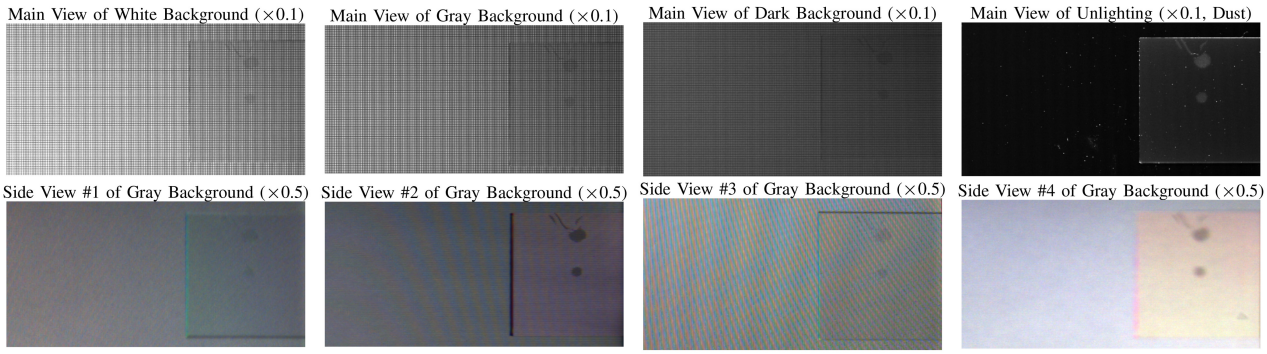


Fig. 6. Overview of the whole captured images from one LCD panel.

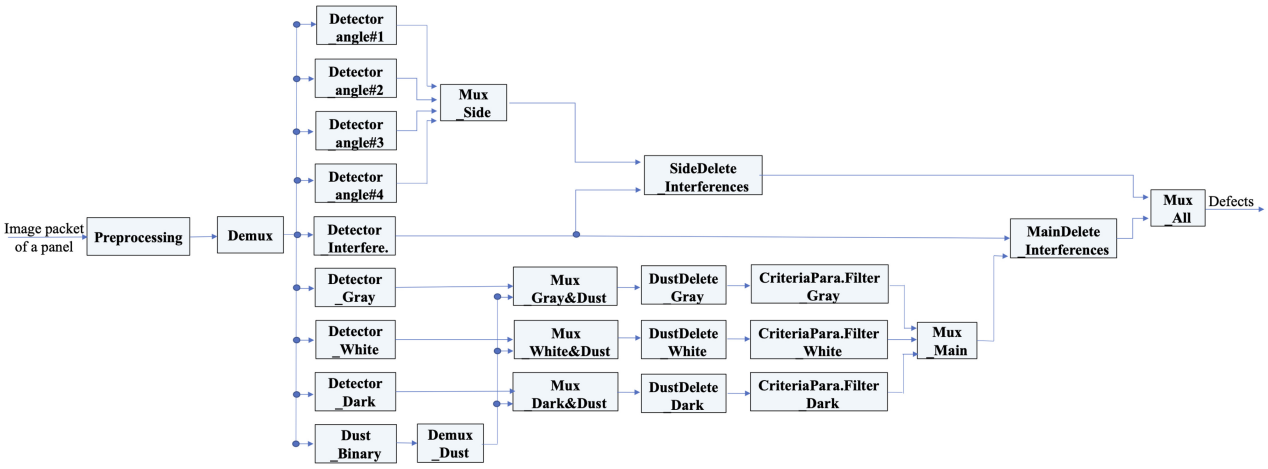


Fig. 7. Software algorithm framework.

It will be resized to 416×416 for further detection described in Section III-C. However, some small defects may be divided into two adjacent subimages (the size of subdefect is less than the area demanded by the defect criteria). It will cause miss alarm. A little overlap with adjacent subimages is necessary when sliding the window. Specifically, the overlap in the horizontal direction is 72 and the vertical direction is 55.

C. Basic Defect Detection Unit

Under the premise of high accuracy, the efficiency of the network is an important factor to be considered. Due to the requirement of detection under different backgrounds from several angles, multiple detectors are designed specially for the defect data sets to ensure the detection performance and work in parallel to obtain a high computational efficiency as shown in Fig. 7. In addition, the number of defects in one panel is normally small. Utilizing two-stage networks as the detector unit, which is represented as Detector_XXX in Fig. 7, will bring unnecessary resource and runtime consumption. It is not necessary to generate too much dense proposals. Numerous interferences are eliminated by special design described above. Hence, a very complex single detector unit is not necessary. A satisfying detection outcome is still allowed to be achieved.

YOLOV3 [25] is employed in this article for its high efficiency. Each image is divided into $S \times S$ grids. If the center position of an object falls in one grid, the grid is responsible for detecting the target. Each grid predicts B bounding boxes, their confidence scores, and the category information. The information of each bounding box contains five data values (x, y, w, h, and confidence score, where x and y refer to the coordinates of the center position of the bounding box predicted by the current grid, and w and h indicate the width and height of the bounding box). The definition of confidence score is shown in (1), where P_r is an abbreviation for precision. When the target is in the grid, $P_r(\text{Object}) = 1$. Otherwise, it is equal to zero. $\text{IoU}_{\text{pred}}^{\text{truth}}$ means the confidence level of the box that includes the whole object. The same object may be detected by multiple bounding boxes. Nonmaximum suppression (NMS) [40] is used to remove redundant boxes, which are inferior

$$\begin{aligned} \text{Confidence} &= P_r(\text{Object}) \times \text{IoU}_{\text{pred}}^{\text{truth}} \\ P_r(\text{Object}) &\in [0, 1]. \end{aligned} \tag{1}$$

D. Interferences Elimination

As is known to all, industrial production involves a wide variety of products varying in size, shape, appearance, packaging, etc. Specifically, as shown in Fig. 9, some kinds of LCD panels are fabricated with several kinds of interferences attached on the protective film or embedded in the panel itself. In order to avoid the false alarm caused by these interferences, one interference detector based on YOLOV3 is employed to detect the positions of the film handle and seal. Then, the detected interferences are subtracted from the results obtained

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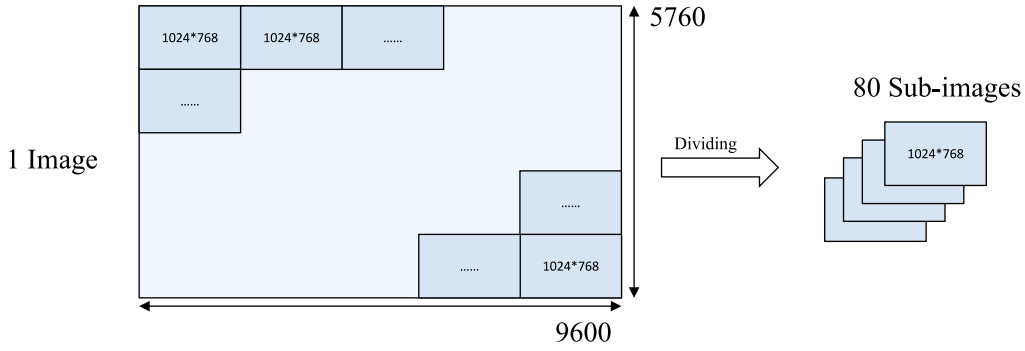


Fig. 8. Scheme of sliding window.

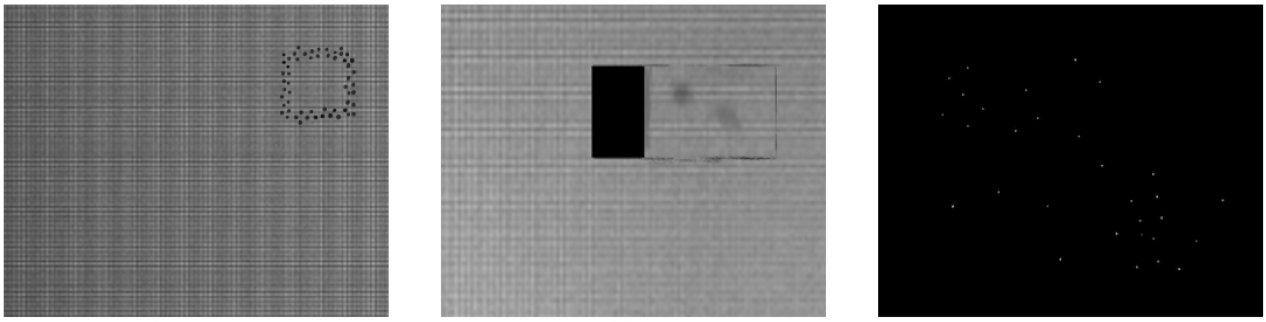


Fig. 9. Interference appearances. (a) Seal embedded in the panel. (b) Film handle attached on the panel surface. (c) Dust adsorbed on the panel surface.

TABLE II
INITIALIZATION PARAMETERS OF YOLOV3

Size of Input Image	Batch Size	Momentum	Learning Rate (LR)	LR Scheduler Mode	Training Epoch
416*416	32	0.9	0.0001	step	100

by the defect detectors. In addition, since the dust can only be captured from the main view, the dust is detected on the power-off image from the main view camera. If the pixel value is greater than the set threshold, it is dust. It can be further subtracted from the detected results.

IV. EXPERIMENTS

YOLOV3 was built by Keras with TensorFlow as its backend. The model we proposed was trained on a standard GPU (NVIDIA Tesla K80) and deployed on the NVIDIA Tesla M60, respectively. Both environments were built on the Ubuntu 16.04 operating system. The initialization parameters of the networks are shown in Table II.

A. Evaluation Metrics

In order to evaluate the performance of our proposed method, computational efficiency of the models is measured by the inference time (seconds). The detection accuracy is measured in terms of precision, recall, and false alarm rate. According to the combination of the ground truth and the predictive class, results can be divided into four types: 1) true positive (TP); 2) false positive (FP); 3) true negative (TN); and 4) false negative (FN). The confusion matrix of the results is shown in Table IV. Precision defined in (2) represents the ability to identify related objects. Recall defined in (3) stands for the ability to find all relevant objects. False alarm rate, generally abbreviated FAR, defined in (4) is known as the probability of false detection

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

$$\text{FAR} = \frac{\text{FP}}{\text{FP} + \text{TN}}. \tag{4}$$

The detection speed is regarded as an index to measure the production capacity and used to compare the business value of different solution. In this article, execution time is calculated as the period starting from capturing the 1st image and ending with achieving the final results. The detection time of manual is regarded as the range from the operator starts to inspect the image and ends with making the decision.

B. Data Set

In order to collect the corresponding data set, we spend three months in one LCD module factory to collect the defective panel from real production lines. According to our statistics, 3000 defective panels involving 53 types of defects were collected. The complexity and authenticity of samples can be

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TABLE III
SAMPLE DISTRIBUTION OF TRAINING DATA SET

Defect Code	The Number of Samples	Cumulative Rate	Visible Background	Visible Angle	Appearance
Type #1	1200	0.34	Gray/White	Main	
Type #2	431	0.47	Dark	Main	
Type #3	222	0.53	Dark	Main	
Type #4	199	0.59	Dark	Main	
Type #5	180	0.64	Gray	Main	
Type #6	179	0.69	Dark/Gray	Main	
Type #7	168	0.74	Gray/White	Main	
Type #8	153	0.78	Dark/Gray	Main	
Type #9	132	0.82	Gray	Side	
Type #10	125	0.85	Gray	Side	
Type #11	120	0.89	Gray	Main/Side	
Type #12	111	0.92	Gray	Main	
Others	280	1	-	-	-

TABLE IV
PERFORMANCE COMPARISON

Method	Time Cost	Labor Cost	Precision	Recall
AOI	30 s/panel	1 person/machine	49/100=0.49	49/50=0.98
Manual Operation	1 min/panel	1 person/machine	48/60=0.8	48/50=0.96

guaranteed. Some of the glimpses of the complexities of the data set can be seen in Figs. 2 and 9. The sample distribution was shown in Table III.

According to Table III, it is obviously observed that defects inspection is an imbalance detection task due to the characteristics of the defects. Hence, data of some certain types with fewer samples are augmented.

C. Performance Comparison

In order to evaluate the effectiveness of our proposed system, it is compared with the traditional manual operation. Experiments involving 5000 LCD panels (including 50 defective panels) are conducted. Three inspection operators are employed and their arbitration result is regarded as the actual performance of manual operation. Through our experiments, the AOI system we proposed reports 100 positives in which 49 alarms are the real defects. The manual operation claims 60 defects and 48 of them are the actual defects. Specifically, the detailed comparison is shown in Table IV. As shown in Fig. 10, it illustrates the visual detection analysis to further investigate the effect of our AOI system.

The results in Table IV illustrate that our framework outperforms manual operation by a large proportion in terms of execution time. It can largely improve the production capacity. Currently, the AOI system still needs one worker to monitor the machine status and help loading and unloading the LCD panel. Nevertheless, the labor can be released when the AOI system is mature and integrated with the mechanical arm to load and unload the LCD panel. Although the precision of AOI is a little lower than the manual scheme, it is not a critical index since it will not degrade the product quality and the

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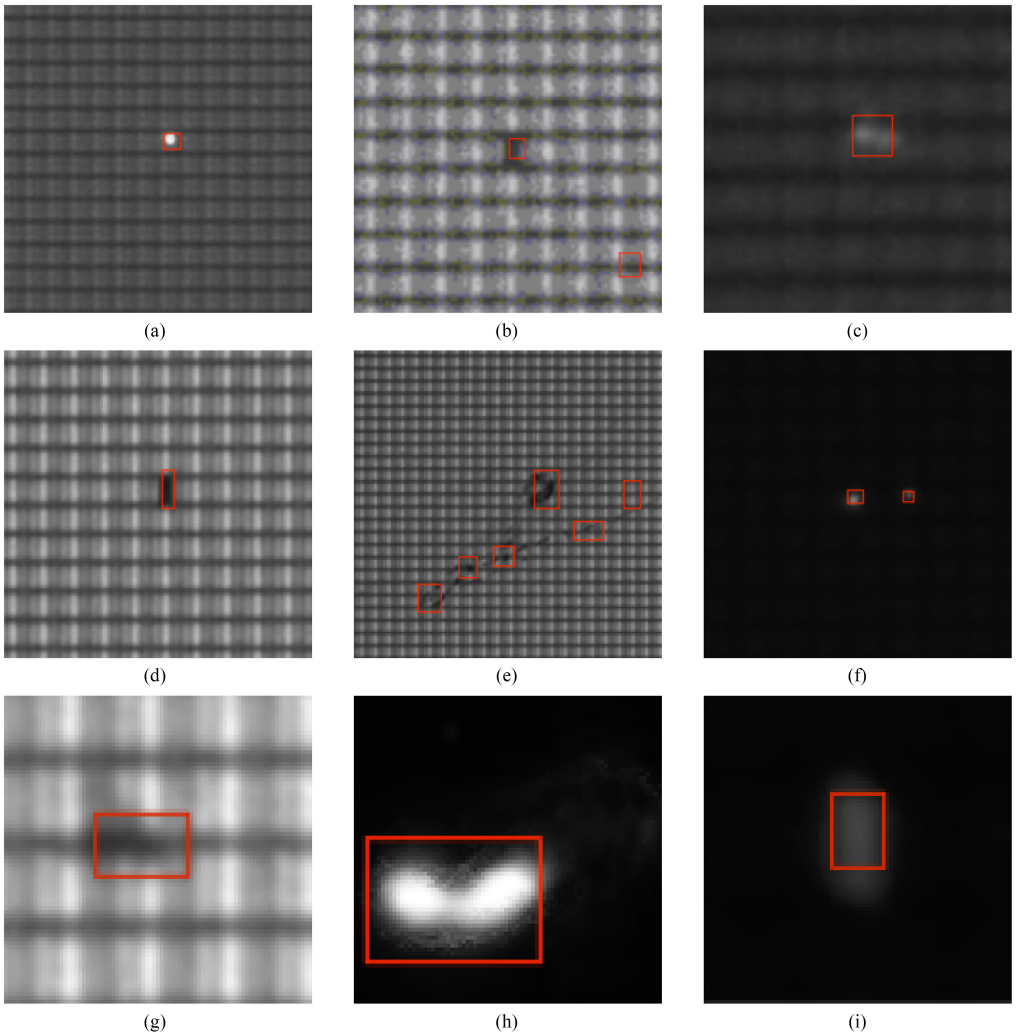


Fig. 10. Examples of detection results with our AOI system.

false alarm rate [51 false alarms in (5000–50)] is acceptable. Even if the panels detected as defective by the AOI system are checked again manually to achieve a higher precision (>80%), it still can save approximately 50% time cost. In addition, the missed alarm is the most concerned thing for all manufacturers. Therefore, the most important index in Table IV is recall, which directly reflects the capability of recognizing the defects. Fortunately, the recall of the AOI system is better than the manual solution, which means the AOI system proposed in this article is practical for defects detection on the module level.

D. Ablation Analysis

In this article, we conducted extensive ablation tests. Initially, for the types of background screen, some colored backgrounds are attempted to be employed. The cost of time rised considerably and the effect is not much better, which is hardly cost effective. In addition, four side view gray cameras with a low cost are adopted by substituting the four side view color cameras. The accuracy declines more seriously. Therefore, the combination of a high-resolution gray camera and four color cameras is favorable. What is more, for the captured images, four side view cameras not only collect gray background images but also the others background images. However, the computational complexity is verified to be extremely high. About the basic defect detection unit, which is represented as Detector_XXX in Fig. 7, two-stage detector, is also be attempted. As expected, it is overwhelmingly time consuming and not necessary. In order to further verify the effectiveness of the detection framework and refine it with the most reasonable convolutional neural network detection unit, which is represented as Detector_XXX in Fig. 7, the performance of the AOI system with Retinanet [29], SSD [27], DSSD [28], and YOLOV3 [25] networks as the convolutional neural network detection unit, respectively, in the detection framework are also compared by experiments. In addition, a single convolutional network, YOLOV3, is directly employed on the original image for further comparison and the effectiveness of our AOI system is further proved. Evaluation metrics are described in Section IV-A.

As shown in Table V, one sees the obvious gain brought by our AOI system in terms of 14.2 times lower false alarm rate, compared with the single network YOLOV3. In addition,

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TABLE V
PERFORMANCE COMPARISON. EACH OF THE AOI MODELS IS ADAPTED BY ADJUSTING THE EMPLOYMENT OF THE CONVOLUTIONAL NEURAL NETWORK DETECTION UNIT FOR DETECTION AND CLASSIFICATION

Method	Time Cost	Labor Cost	Precision	Recall	False Alarm Rate
AOI with YOLOV3	30 s/panel	1 person/machine	49/100=0.49	49/50=0.98	(100-49)/(5000-50)=0.010
AOI with SSD	32.7 s/panel	1 person/machine	48/118=0.41	48/50=0.96	(118-48)/(5000-50)=0.014
AOI with DSSD	34.4 s/panel	1 person/machine	49/107=0.46	49/50=0.98	(107-49)/(5000-50)=0.012
AOI with Retinanet	33.4 s/panel	1 person/machine	48/109=0.44	48/50=0.96	(109-48)/(5000-50)=0.012
YOLOV3	5.43 s/panel	1 person/machine	9/712=0.01	9/50=0.18	(712-9)/(5000-50)=0.142

the AOI system with YOLOV3 achieved superior performance in terms of both the detection accuracy and computational efficiency compared to other convolutional neural networks. Hence, YOLOV3 is a better choice as the convolutional neural network detection unit in the real application and employed in our AOI system after the experimental analysis.

For the model composition, different detectors are tried to be assigned to different cameras. Although fewer detectors can be used, the accuracy is not acceptable because different defects cannot be distinguished well in advance by different backgrounds and angles. Finally, in the system, we omit different components to explore the effectiveness of each component, including DustDelete_XXX, XXX_Interferences, and CriteriaPara.Filter_XXX in Fig. 7. After extensive experiments, the system shown in Fig. 7 shows great advantages in terms of the runtime and detection accuracy.

V. DISCUSSION

Defects detection equipped with artificial intelligence plays a considerably important role in the development of Industry 4.0, which is a new production method and inevitable trend of industrial development. The AOI system is the core of an IoT paradigm for defect detection. Intelligent machines (systems) will replace most of the human physical labor and a considerable part of the mental labor. It not only allows human to spend more time on creative work but also reduces the consumption and waste of resources and energy. The manufacturing industry is leaded to a sustainable development.

Specifically, the AOI system for defects detection we proposed is described above in detail. It can be employed to work in an independent node on the production line in a factory. It is hard to collect data efficiently, update the model online, and provide global macrostatistics for the optimization of the production process. As an IoT paradigm application, more valuable is that the AOI systems are utilized in a distributed integrated system to achieve a better automatic quality monitoring. They are the key components of the intelligence inspection system shown in Fig. 11. The benefits include facilitating dynamic training and updating models, saving GPU computing resources, and providing data support for various factory decisions. Such monitoring technology in the distributed integrated inspection system is capable to strengthen the information management and services, which can arrange production plans more reasonably. Specifically, compared with manual detection technologies, its advantages mainly include the following.

1) High degree of intelligence, with consistency and repeatability unmatched by humans.
2) Fast detection speed, high accuracy, and low missed detection rate.
3) Machine vision and intelligent control technology are combined to achieve vision-based high-speed defects detection.
4) Strengthening information management to arrange production plans more reasonably.

As shown in Fig. 11 (an IoT paradigm), smart factories (SF) are dispatched by the intelligent control center through the wide-area network (WAN). Factories are linked by each other in the paradigm. Information is disseminated in the IoT paradigm for inspection management, such as model updating, images collecting, etc. Specifically, the job scheduler in each SF is responsible for task assignment, receiving the inspection results from the AOI systems, making decision analysis and monitoring the AOI system to prevent no response and reporting anomalies with real-time communication using the local area network (LAN). Moreover, image collection module is responsible to collect images used to train and update the AOI systems and generate a database as well. In addition, intelligent inspection relies on the cloud and fog (edge computing) collaboration to deal with massive data. Cloud computing is responsible for big data analysis of nonreal-time and long-period data. It plays a special role in model training. Edge computing focuses on the analysis of real-time and short-period data. It can better support real-time intelligent defects detection and provide the necessary edge device data for the cloud. The combination of edge computing and cloud computing provides a more perfect data processing platform for intelligent inspection. Basically, the AOI system consists of imaging system and defects detection framework. Every component plays an important role in the system. In summary, the AOI system is significant for intelligent inspection with IoT. Vigorously developing the intelligent inspection system is of great significance for upgrading of the manufacturing industry, improving production efficiency, reducing energy and resources consumption, and achieving intelligent and green development of the manufacturing process. Combined with other emerging technologies, such as robotics and IoT, a high-efficiency, energy-saving, and environmentally friendly smart factory that is pivotal for Industry 4.0 can be achievable.

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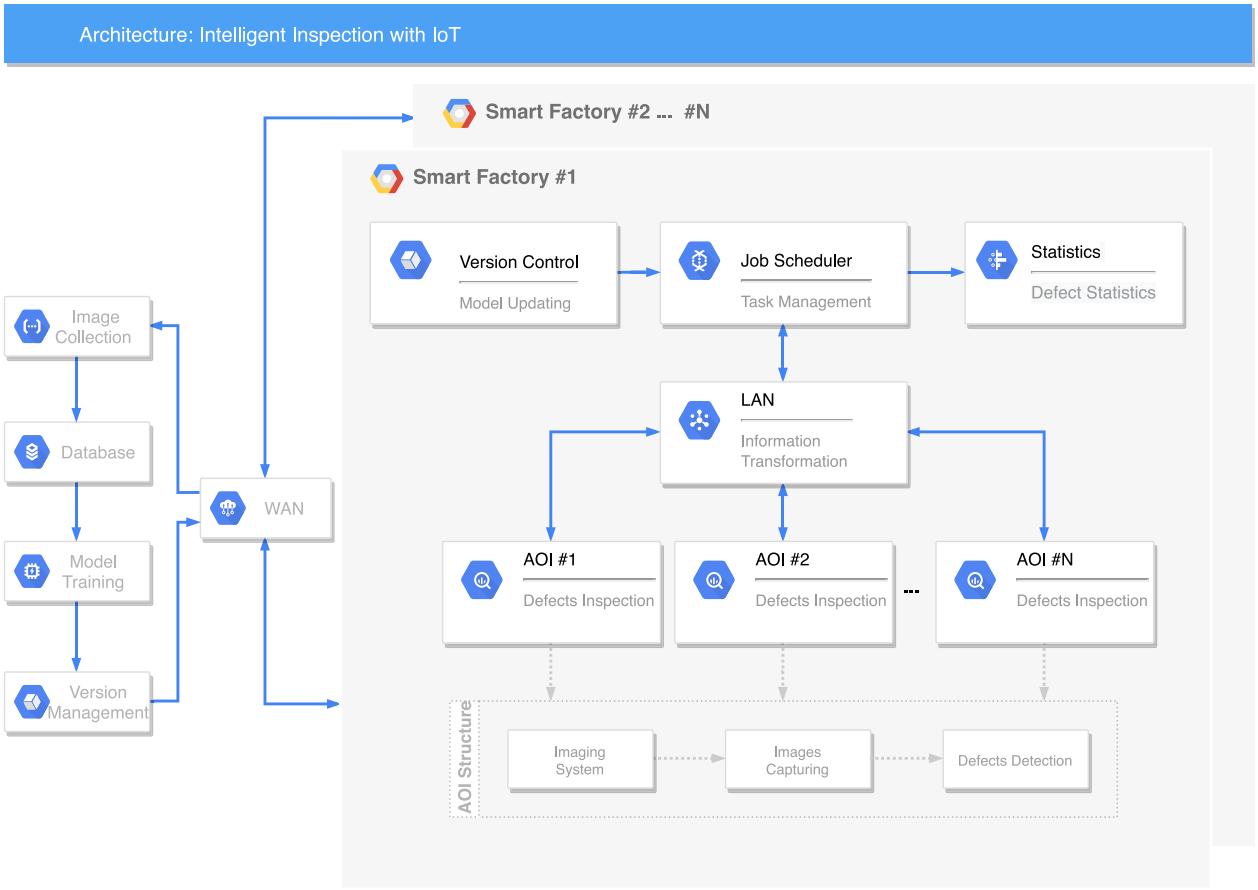


Fig. 11. Entire scheme architecture.

VI. Conclusion

This study develops a practical AOI system for module-level defects detection. As an application of the IoT paradigm, it is discussed in detail in Section V. It has been deployed in a real-world LCD manufacturing line from a major player in the world.

The LCD panel is triggered to display multiple background images to build diverse contrast environments to reveal defects as more as possible. Considering various visible angles of the defects, multiple cameras are conducted to capture the background scenes from different angles. In order to increase the detection accuracy, we proposed a combined detection framework based on YOLOV3. The detected defects are refined by removing the interferences induced by the dust and protective film damages. According to the CriteriaPara.Filter_XXX modules, various defects can be detected. Examined by extensive experiments, the AOI system outperformed the manual solution by a large margin. Specifically, it achieves 2.0 times faster speed than the manual operation and brings a gain of 2.08% in terms of the recall. In summary, a distributed intelligent inspection system with the capabilities of automatically collecting defective images is designed and implemented in a well-known manufacturing enterprises.

In the future, more work will be considered to focus on applying manipulators in manufacturing enterprises. It can further save the labor resources. In addition, more attention will be put to apply our proposed system in more factories. Establishing smart factory with AOI systems in a wider scope is further considered, which can promote the development of industry 4.0 significantly.

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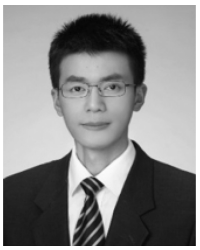
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